

Progress Report

Project scope update

The original project proposal focused on analyzing the relationship between product price, customer ratings, and review counts for Amazon electronics-related products. The overall research direction above remained the same, but the data collection plan was expanded. In addition to the two original CSV datasets, I added an API-based data source and collected product information from three Amazon Best Sellers webpages.

Data Sources

The project uses four data sources. The first two are `All_Electronics.csv` and `Home_Entertainment_Systems.csv`, which contain product name, category, rating, number of ratings, and price-related variables. The third source is product data collected from three categories of Amazon Best Sellers webpages, including Electronics, Cell Phones & Accessories, and Computers & Accessories. These webpages were scraped and merged into one combined CSV file. The fourth source is a product dataset obtained through the DummyJSON API. This API-based dataset includes product name, price, rating, review count, category, and source information. All four datasets were loaded, cleaned, and standardized into a unified schema with the following fields: `product_name`, `price`, `rating`, `review_count`, `category`, `subcategory`, and `source`. After cleaning and combining the datasets, the final merged dataset contains 18,016 observations.

The project currently uses the DummyJSON Products API as an API-based data source. The API was accessed in Python using the `requests` library, and the returned JSON data was converted into a pandas DataFrame. Relevant fields were extracted and renamed to match the project schema.

Issues / difficulties

The first major difficulty was that the different data sources used different column names and formats, so I cleaned and standardized these data sources before combining them. Another challenge was webpage scraping from Amazon Best Sellers pages, because webpage structure can be inconsistent and may limit the number of accessible product records per page. At the current stage, the scraper successfully collected data from three pages, but each page returned fewer records than the expected top 100 items. In the next stage, I plan to further improve data collection and continue exploratory analysis of the relationships among price, rating, and review count.